

Sno	Author Names	Title of the Paper
1	Sharada P. Mohanty; David P. Hughes; Marcel Salathé	Using Deep Learning for Image-Based Plant Disease Detection
2	Guan Wang; Yu Sun; Jianxin Wang	Automatic Image-Based Plant Disease Severity Estimation Using Deep Learning
3	Amanda Ramcharan; Kelsee Baranowski; Peter McCloskey; Babuali Ahmed; James Legg; David P. Hughes	Deep Learning for Image-Based Cassava Disease Detection
4	H. A. Atabay	Deep Residual Learning For Tomato Plant Leaf Disease Identification
5	Justine Boulent; Sylvie Foucher; Jérôme Théau; Pierre-Luc St-Charles	Convolutional Neural Networks for the Automatic Identification of Plant Diseases
6	Sue Han Lee; Hervé Goëau; Pierre Bonnet; Alexis Joly	Attention-Based Recurrent Neural Network for Plant Disease Classification
7	Junde Chen; Jinxiu Chen; Defu Zhang; Yuandong Sun; Yaser A. Nanekaran	Using Deep Transfer Learning for Image-Based Plant Disease Identification
8	Ümit Atila; Murat Uçar; Kemal Akyol; Emine Uçar	Plant Leaf Disease Classification Using EfficientNet Deep Learning Model

9	Penghui Gui; Wenjie Dang; Feiyu Zhu; Qijun Zhao	Towards Automatic Field Plant Disease Recognition
10	Waleed Albattah; Marriam Nawaz; Ali Javed; Momina Masood; Saleh Albahli et al.	A Novel Deep Learning Method for Detection and Classification of Plant Diseases
11	Antonio Bruno; Davide Moroni; Riccardo Dainelli; Leandro Rocchi; Silvia Morelli; Emilio Ferrari; Piero Toscano; Massimo Martinelli	Improving Plant Disease Classification by Adaptive Minimal Ensembling
12	Riyao Chen; Haixia Qi; Yu Liang; Mingchao Yang	Identification of Plant Leaf Diseases by Deep Learning Based on Channel Attention and Channel Pruning
13	Andrew J.; Jennifer Eunice; Jude Hemanth et al.	Deep Learning-Based Leaf Disease Detection in Crops Using Images for Agricultural Applications
14	Divakar Yadav; Aarushi Gupta; Arti Jain; Arun Kumar Yadav	Plant Leaf Disease Detection Using CNN with Transfer Learning and XGBoost
15	Amitava Nag; Priya Raj Chanda; Sukumar Nandi	Mobile App-Based Tomato Disease Identification with Fine-Tuned Convolutional Neural Networks
16	Authors of the IEEE ICSSIT study	Plant Disease Classification Using CNN-LSTM Techniques
17	M. Shafik et al.	Using Transfer Learning-Based Plant Disease Classification and Detection for Sustainable Agriculture

18	Geetabai S. Hukkeri; B. C. Soundarya; H. L. Gururaj; Vinayakumar Ravi	Classification of Various Plant Leaf Disease Using Pretrained Convolutional Neural Network on ImageNet
19	S. Karthikeyan; R. Charan; Sathiya Narayanan; L. Jani Anbarasi	Enhanced Plant Disease Classification with Attention-Based Convolutional Neural Network Using Squeeze and Excitation Mechanism
20	Zafar Salman; Abdullah Muhammad; Dongil Han	Plant Disease Classification in the Wild Using Vision Transformers and Mixture of Experts
21	Muhammad Hammad Saleem; Johan Potgieter; Khalid Mahmood Arif	A Weight Optimization-Based Transfer Learning Approach for Plant Disease Detection of New Zealand Vegetables
22	Hasibul Islam Peyal; Md. Nahiduzzaman; Md. Abu Hanif Pramanik; Md. Khalid Syfullah; Saleh Mohammed Shahriar; Abida Sultana; Mominul Ahsan; Julfikar Haider; Amith Khandakar; Muhammad	Plant Disease Classifier: Detection of Dual-Crop Diseases Using Lightweight 2D CNN Architecture
23	Sana Parez; Naqqash Dilshad; Turki M. Alanazi; Jong Weon Lee	Towards Sustainable Agricultural Systems: A Lightweight Deep Learning Model for Plant Disease Detection
24	Guoqiang Li; Yuchao Wang; Qing Zhao; Peiyan Yuan; Baofang Chang	PMVT: A Lightweight Vision Transformer for Plant Disease Identification on Mobile Devices
25	Xiaotian Wang; Weiqun Cao	GACN: Generative Adversarial Classified Network for Balancing Plant Disease Dataset and Plant Disease Recognition

Year of Published	Dataset Used
2016	PlantVillage – 54,306 images; 14 crops; 38 crop-disease classes
2017	PlantVillage apple black rot images; 4 severity stages
2017	Cassava leaf disease images collected in Tanzania
2017	Tomato leaf disease images
2019	Review of 19 CNN-based plant disease studies and their datasets
2020	PlantVillage; 21 disease/healthy classes derived from the dataset
2020	Public plant disease dataset and rice plant images under complex backgrounds
2021	PlantVillage – 55,448 original images; 39 classes; augmented version also evaluated

2021	PlantVillage for training; Field-PlantVillage (665 field images) for independent testing
2022	PlantVillage Kaggle database
2022	PlantVillage original and augmented datasets
2022	PlantVillage; local peanut leaf disease dataset
2022	PlantVillage – 38 classes
2023	PlantVillage
2023	PlantVillage tomato images plus self-captured tomato leaf images
2023	PlantVillage
2024	PlantVillage – 15 selected classes / 54,303 images

2024	PlantVillage – 38 classes and 14 plant species
2025	PlantVillage – 54,305 images; 38 classes; additional potato dataset
2025	PlantVillage and PlantDoc
2022	NZDLPlantDisease-v2 – 28 healthy/defective organs across 8 vegetables; agricultural transfer weights
2023	PlantVillage and dual-crop leaf disease images
2023	PlantVillage, Data Repository of Leaf Images (DRLI), and Plant Composite dataset
2023	Wheat, coffee and rice plant disease datasets
2023	PlantVillage and AI Challenger 2018

Methodology Used
Deep CNNs (AlexNet and GoogLeNet) for image classification
CNNs trained from scratch and VGG16 with transfer learning
Inception V3 transfer learning / deep CNN classification
Residual deep learning / ResNet-based classification
Literature review of CNN approaches for crop disease identification
GoogLeNet CNN feature extraction with attention-based RNN
ImageNet-pretrained VGGNet combined with Inception module (INC-VGGN) using transfer learning
EfficientNet B4/B5 and other pretrained CNNs using transfer learning

Improved CNN with background replacement, leaf resizing, channel orthogonal constraint and auxiliary species classification
Custom CenterNet with DenseNet-77 backbone, transfer learning and annotated regions
EfficientNet-B0 transfer learning, regularization, advanced optimization and adaptive feature-level ensemble
ResNet-18 with channel attention (CACPNET) and L1-norm channel pruning
Comparison of VGG16, InceptionV4, ResNet50 and DenseNet121 with deep learning/transfer learning
ResNet50 transfer learning for feature extraction followed by XGBoost classification
Transfer learning with AlexNet, ResNet50, SqueezeNet-1.1, VGG19 and DenseNet121; data augmentation; mobile app
Transfer-learning feature extraction followed by an LSTM classifier (CNN-LSTM hybrid)
Nine pretrained CNNs, deep feature extraction, fine-tuning, early-fusion ensemble and lead-voting ensemble

Transfer learning with VGG16, ResNet50, InceptionV3, MobileNetV2, AlexNet and EfficientNet

Custom CNN with Squeeze-and-Excitation identity blocks (CNN-SEEIB)

Vision Transformer backbone with Mixture of Experts, entropy regularization and orthogonal regularization

Transfer learning with weights optimized from agricultural datasets; Faster R-CNN based evaluation

Lightweight 2D CNN with Adam optimizer and categorical cross-entropy

E-GreenNet using MobileNetV3Small backbone with lightweight deep learning

MobileViT-based PMVT with inverted residual structure and CBAM attention

Generative Adversarial Classified Network with generator, discriminator and classifier for synthetic-image augmentation

Limitation of the Paper

High performance on controlled images, but accuracy dropped strongly on images from different real-world conditions.

Focused on one disease (apple black rot) and severity estimation; broader crop/disease generalization was not demonstrated.

Limited to cassava and a specific geographic field setting; broader crop and environmental variation requires further validation.

Focused on tomato diseases; performance under diverse field conditions was not fully established.

It is a survey rather than a new experimental model, so it does not provide a single unified dataset or deployment solution.

PlantVillage is controlled; cross-dataset/field generalization remains challenging.

Performance on complex backgrounds was lower than controlled conditions, and the approach increases model complexity.

Very high performance was obtained on PlantVillage, but the dataset is largely controlled and may not represent field conditions.

Field performance remained substantially lower than PlantVillage performance, showing a domain/generalization gap.
Requires region annotations and a more complex detection pipeline; real-world multi-disease deployment remains challenging.
Ensembling adds implementation complexity and was evaluated mainly on the benchmark dataset.
Additional attention/pruning design increases methodological complexity; broader field validation is still needed.
Performance was demonstrated on PlantVillage; controlled images limit conclusions about real-world field generalization.
Hybrid CNN-XGBoost pipeline is more complex than an end-to-end CNN and requires additional classifier tuning.
Self-collected data was geographically limited to northeastern India, and disease coverage is restricted to the selected tomato classes.
The hybrid architecture is more complex than a standard CNN and was evaluated on the benchmark dataset rather than broad field imagery.
The study used a selected 15-class subset; class imbalance and real-time field data remain important challenges.

Experiments use benchmark PlantVillage images; real-world environmental variability and deployment constraints are not fully addressed.

Custom architecture requires additional design/tuning and still needs broader field validation.

Cross-domain accuracy remains much lower than controlled PlantVillage performance, showing persistent domain-shift challenges.

Dataset is specific to New Zealand vegetables and requires further testing across crops, regions and environmental conditions.

The model is focused on selected crops/diseases; broader cross-domain validation is still required.

Performance varies across datasets, and generalization to unseen field conditions remains a challenge.

Focuses on selected crops and mobile deployment; it does not directly establish performance across the full PlantVillage 38-class problem.

Synthetic-image generation quality is still limited at higher resolutions; GAN training also increases computational complexity.